Silverio Martínez Andrés Grupo 3S Tarea 28

Nombre:

Tema:

Día | Mes | Año | Folio

Cal. 10

Tarea 28

1.- Calcular las derivadas de orden superior de la función

$$F(x) = \frac{x^4}{4} - \frac{4x^3}{3} + 5x - 1$$

$$F'(x) = x^3 - 4x^2 + 5$$

$$F''(x) = 3x^2 - 8x$$

$$F'''(x) = 6x - 8$$

$$F^{IV}(x) = 6$$

$$F^{V}(x) = 0$$

2.- Sea $F(x) = x^{1/3}$

Obtener el valor de la segunda derivada para $x = 8$

$$F'(x) = \frac{1}{3}x^{-2/3}$$

$$F''(x) = -\frac{2}{9}x^{-5/3} = -\frac{2}{9x^{5/3}} = -\frac{2}{9\sqrt[3]{x^5}} = -\frac{2}{9\sqrt[3]{x^3}\sqrt[3]{x^2}}$$

$$F''(x) = \frac{-2}{9x\sqrt[3]{x^2}}$$

$$F''(8) = -\frac{-2}{9(8)\sqrt[3]{(8)^2}} = \frac{-2}{9(8)4} = \frac{-1}{144}$$

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3.- Determinar la segunda derivada de $xy - y^2 + 2 = 0$

$$xy - y^2 + 2 = 0$$

$$xy' + y - 2yy' = 0$$

$$xy' - 2yy' = -y$$

$$(x - 2y)y' = -y$$

$$y' = \frac{-y}{x-2y}$$

$$y' = \frac{(x-2y)(-y') - (-y)(1-2y')}{(x-2y)^2}$$

$$y' = \frac{-xy' + 2yy' + y - 2yy'}{(x-2y)^2}$$

$$y'' = \frac{y - xy'}{(x-2y)^2}$$

$$y'' = \frac{y - \frac{x}{1}\left(\frac{-y}{x-2y}\right)}{(x-2y)^2}$$

$$y'' = \frac{\frac{y}{1} + \frac{xy}{x-2y}}{(x-2y)^2}$$

$$y'' = \frac{\frac{(x-2y)y + xy}{x-2y}}{(x-2y)^2}$$

$$y'' = \frac{\frac{xy - 2y^2 + xy}{x-2y}}{(x-2y)^2}$$

$$y'' = \frac{\frac{2xy - 2y^2}{(x-2y)}}{\frac{(x-2y)^2}{1}}$$

$$y'' = \frac{2xy - 2y^2}{(x-2y)^3}$$

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4.- Obtener $D^3_x y$ de la función $\begin{cases} x = \operatorname{Sen} t \\ y = \cos^2 t \end{cases}$

$$D_x y = \frac{D_t y}{D_t x} = \frac{2\cancel{\cos t}(-\operatorname{sen} t)(1)}{\cancel{\cos t}\,(1)} \text{ R.C. } = -2\operatorname{sen} t$$

$$D^2_x y = \frac{D_t(D_x y)}{D_t x} = \frac{-2\cos t(1)}{\cos t(1)} = -2$$

$$D^3_x y = \frac{D_t(D^2_x y)}{D_t x} = \frac{0}{\cos t(1)} = 0$$

5.- Sea la función $\begin{cases} x = \operatorname{ang} tg\ t^2 \\ y = t^2 + 2 \end{cases}$

Calcular $f'''(t)$ y valuarla para $t = 1$

$$f'(t) = \frac{2t}{\frac{2t}{1+(t^2)^2}} = \frac{\frac{2t}{1}}{\frac{2t}{1+t^4}} = \frac{\cancel{2t}(1+t^4)}{\cancel{2t}} = 1+t^4$$

$$f''(t) = \frac{\frac{4t^3}{1}}{\frac{2t}{1+t^4}} = \frac{\cancel{4t^3}(1+t^4)}{\cancel{2t}} = 2t^2(1+t^4) = 2t^2+2t^6$$

$$f'''(t) = \frac{\frac{4t+12t^5}{1}}{\frac{2t}{1+t^4}} = \frac{(4t+12t^5)(1+t^4)}{2t} = \frac{4t+4t^5+12t^5+12t^9}{2t}$$

$$= 2+2t^4+6t^4+6t^8$$

$$f'''(t) = 2+8t^4+6t^8$$

$$f'''(1) = 2+8(1)^4+6(1)^8$$

$$f'''(1) = 2+8+6 = 16$$